

HTTP REQUESTS

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API DESCRIPTION

An **E-Commerce API Application** is a web-based service that enables online shopping platforms to manage and exchange data between the client (frontend) and server (backend). It provides structured endpoints that allow applications to perform operations such as retrieving product details, adding new products, updating existing information, and deleting products using standard HTTP methods like GET, POST, PUT, PATCH, and DELETE.

In this project, the **Fake Store API** is used as a sample e-commerce backend service. It simulates a real online store by providing product data including title, price, description, category, and images. This API helps demonstrate how different HTTP requests work in real-world e-commerce applications.

Through Postman, the following operations are performed:

- **GET** – Retrieve product details from the server
- **POST** – Add a new product to the store
- **PUT** – Update complete product information
- **PATCH** – Modify specific product fields
- **DELETE** – Remove a product from the store

E-commerce APIs play a crucial role in modern web applications by ensuring smooth communication between the user interface and the database. They make online shopping systems scalable, maintainable, and secure by handling data operations efficiently.

REQUESTS

1.GET

GET is an HTTP method used to **retrieve data from a server**.

It is mainly used to **read or fetch information** without modifying the data on the server.

Key Points

- Used to **fetch data**
- Does **not change** data
- Data is sent through the **URL**
- Safe and idempotent method

- Returns response in **JSON format** (usually)

◆ URL:

<https://fakestoreapi.com/products/1>

◆ METHOD:

GET

◆ DESCRIPTION:

GET method is used to retrieve data from the server without modifying it.

◆ RESPONSE:

- Status Code: 200 OK
- Data returned in JSON format

The screenshot displays the Postman interface within 'Yuva Dharshini's Workspace'. A GET request is configured with the URL 'https://fakestoreapi.com/products/1'. The 'Body' tab is selected, showing a raw JSON response. The status bar indicates a successful '200 OK' response with a time of 1.37s and a size of 954B. The response body is a JSON object containing product details.

```
1 {
2   "userId": 2,
3   "id": 2,
4   "title": "Web Programming",
5   "body": "Postman tool"
6 }
```

Below the request, the 'Body' section shows the received JSON response:

```
1 {
2   "id": 1,
3   "title": "Fjallraven - Foldsack No. 1 Backpack, Fits 15 Laptops",
4   "price": 109.95,
5   "description": "Your perfect pack for everyday use and walks in the forest.
6     Stash your laptop (up to 15 inches) in the padded sleeve, your everyday",
7   "category": "men's clothing",
8   "image": "https://fakestoreapi.com/img/81fPKd-2AYL._AC_SL1500_t.png",
9   "rating": {
10     "rate": 3.9,
11     "count": 120
12   }
13 }
```

2.POST

POST is an HTTP method used to create new data on the server.
It is mainly used to send data to the server to create a new resource.

Key Points

- Used to create new data
- Sends data in request body
- Changes server data
- Not idempotent
- Returns newly created resource

◇ URL:

<https://fakestoreapi.com/products>

◇ METHOD:

POST

◇ REQUEST BODY (JSON):

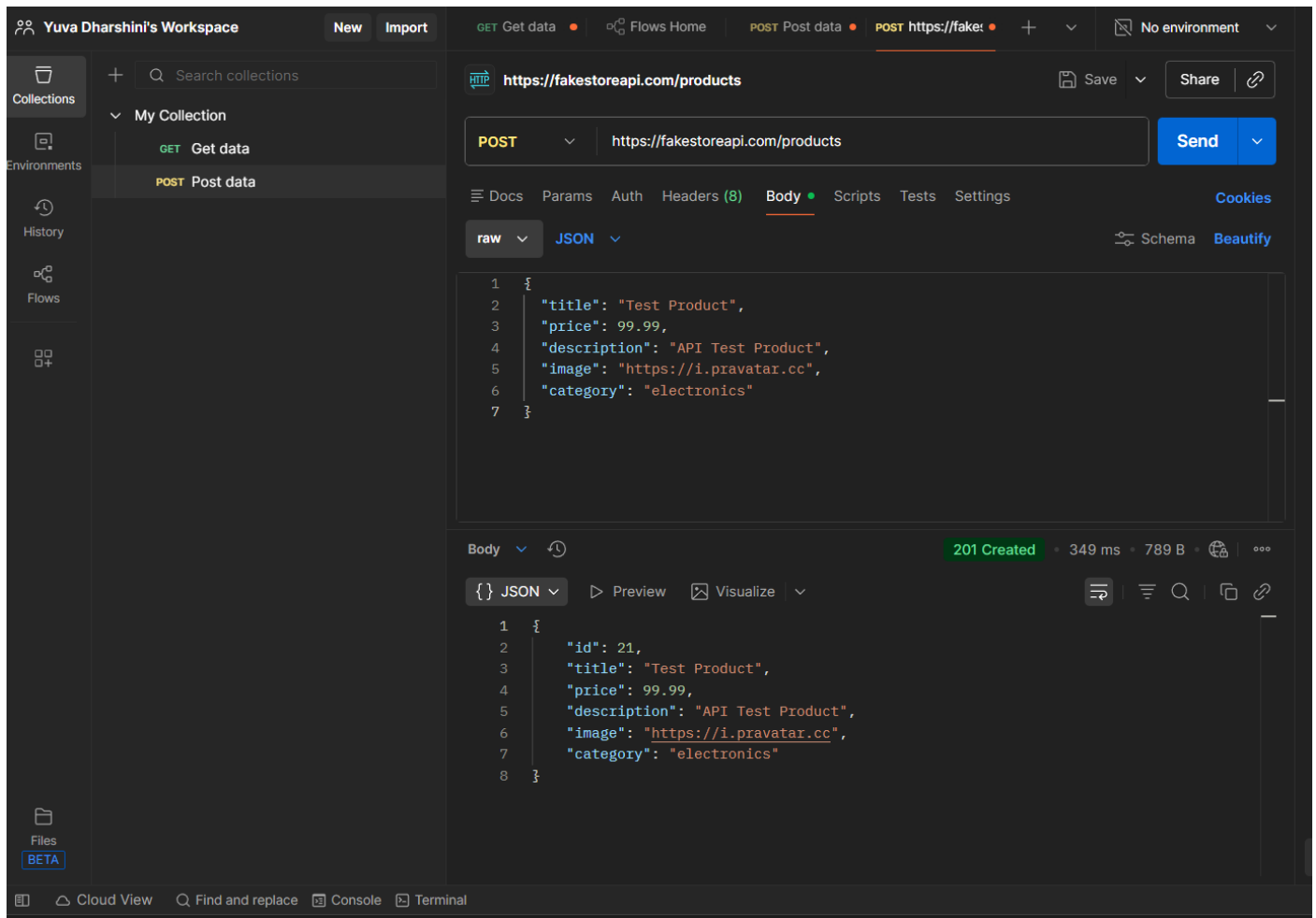
```
{  
  "title": "Web Programming Product",  
  "price": 399.99,  
  "description": "Created using Postman",  
  "image": "https://i.pravatar.cc",  
  "category": "electronics"  
}
```

◇ DESCRIPTION:

POST method is used to create a new resource on the server by sending data in the request body.

◇ RESPONSE:

- Status Code: 200 OK / 201 Created
- Newly created product returned in JSON format



3. PUT

PUT is an HTTP method used to update an existing resource completely on the server. It replaces the entire data of the resource with new data.

Key Points

- Used to update entire resource
- Sends full data in request body
- Idempotent method
- Replaces existing data
- Returns updated resource

◇ URL:

<https://fakestoreapi.com/products/1>

◇ METHOD:

PUT

◇ REQUEST BODY (JSON):

```
{
  "title": "Updated Product Title",
  "price": 499.99,
```

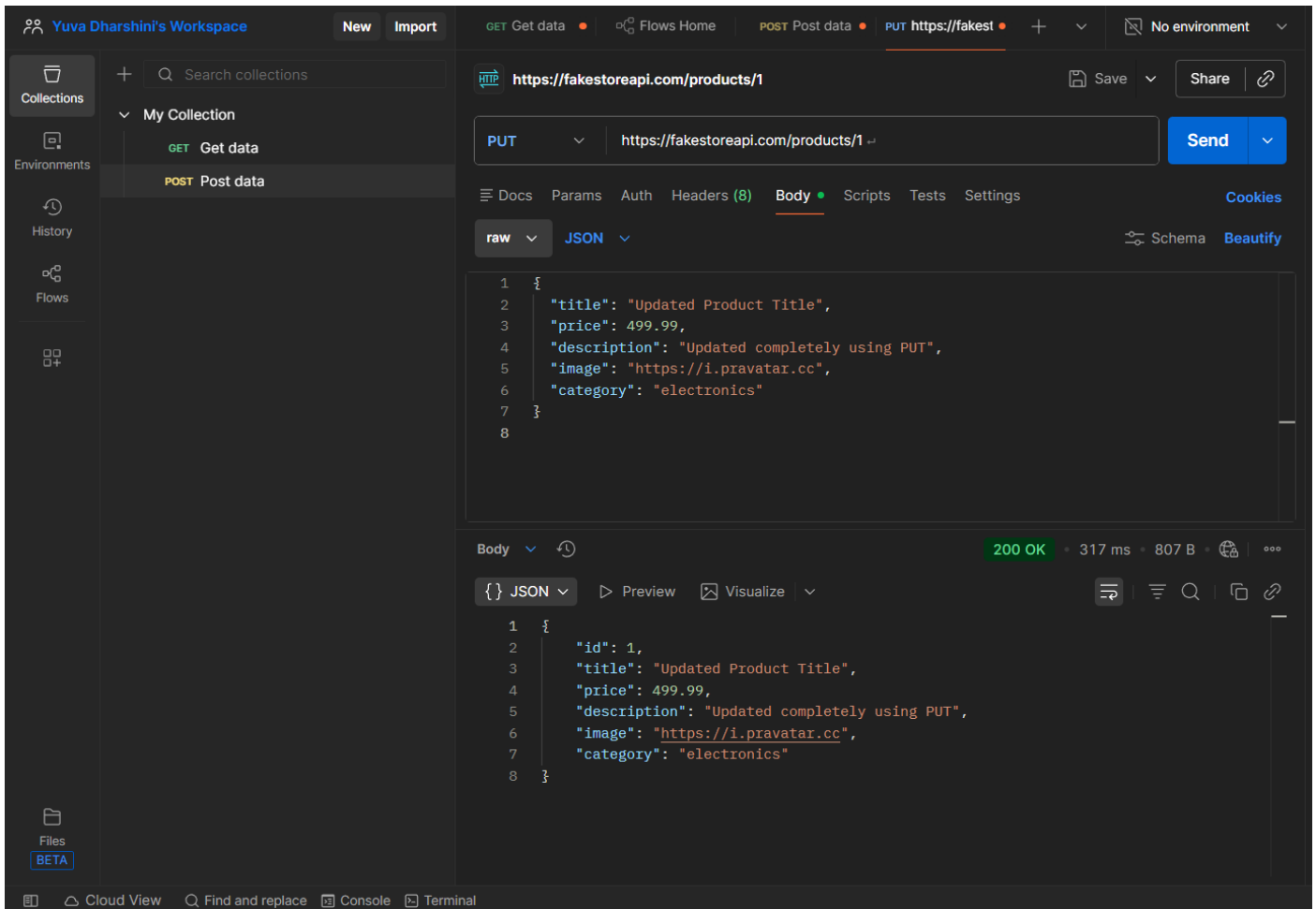
```
}
  "description": "Updated completely using PUT",
  "image": "https://i.pravatar.cc",
  "category": "electronics"
}
```

◆ DESCRIPTION:

PUT method is used to update the entire resource by replacing the existing data with new data.

◆ RESPONSE:

- Status Code: 200 OK
- Updated product returned in JSON format



4.PATCH

PATCH is an HTTP method used to update only specific fields of an existing resource. It modifies partial data without replacing the entire resource.

Key Points

- Used for partial update
- Sends only required fields
- Changes server data

- More efficient than PUT for small updates
- Returns updated resource

◆ **URL:**

<https://fakestoreapi.com/products/1>

◆ **METHOD:**

PATCH

◆ **REQUEST BODY (JSON):**

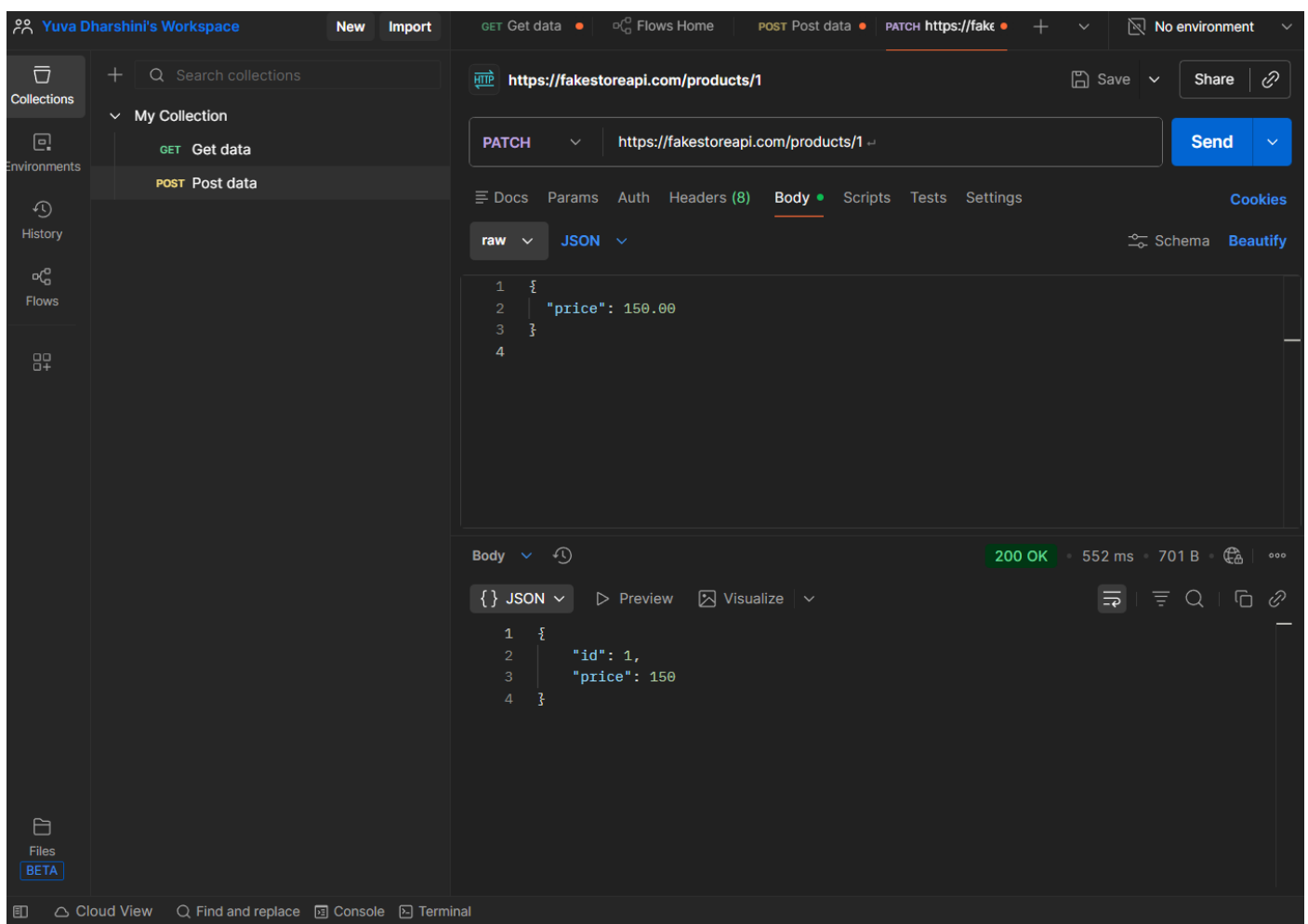
```
{  
  "price": 299.99  
}
```

◆ **DESCRIPTION:**

PATCH method is used to update only specific fields of a resource without affecting other data.

◆ **RESPONSE:**

- Status Code: 200 OK
- Updated product returned in JSON format



5.DELETE

DELETE is an HTTP method used to remove an existing resource from the server.
It permanently deletes the specified data.

Key Points

- Used to delete resource
- Changes server data
- Idempotent method
- Removes specified data
- May return empty response

◇ URL:

<https://fakestoreapi.com/products/1>

◇ METHOD:

DELETE

◇ DESCRIPTION:

DELETE method is used to remove a resource from the server.

◇ RESPONSE:

- Status Code: 200 OK / 204 No Content
- Resource deleted successfully

The screenshot displays a REST client interface with the following components:

- Header:** Yuva Dharshini's Workspace, New, Import, GET Get data, POST Post data, DEL https://fakestoreapi.com/products/1, No environment.
- URL Bar:** https://fakestoreapi.com/products/1, Save, Share.
- Method:** DELETE, Send.
- Body:** none (This request does not have a body).
- Response:** 200 OK, 391 ms, 962 B. The response body is a JSON object:

```
{  "id": 1,  "title": "Fjallraven - Foldsack No. 1 Backpack, Fits 15 Laptops",  "price": 109.95,  "description": "Your perfect pack for everyday use and walks in the forest. Stash your laptop (up to 15 inches) in the padded sleeve, your everyday",  "category": "men's clothing",  "image": "https://fakestoreapi.com/img/81fPKd-2AYL._AC_SL1500_t.png",  "rating": {    "rate": 3.9,    "count": 120  }}
```
- Footer:** Cloud View, Find and replace, Console, Terminal.